4 **(a)** A contractor tries to measure the depth of a new well shaft to build a ladder to reach the bottom of the shaft. He uses an audio oscillator with adjustable frequency and applies it across the top of the well. Two successive resonances are heard at 70.6 Hz and 90.8 Hz. The speed of sound is 343 ms^{-1}. Determine the depth of the well.

depth of well = ……. m [3]

(b) An engineering student designed an Automated Guided Vehicle (AGV) using interference of radio waves from two coherent emitters 2.0 m apart emitting radio waves of frequency *f* that are in phase as shown in Fig. 4.1. The computer on the AGV detects and searches for lines of constructive interference and adjusts the AGV so that it is always aligned along the centre-line, in the middle of the emitters.

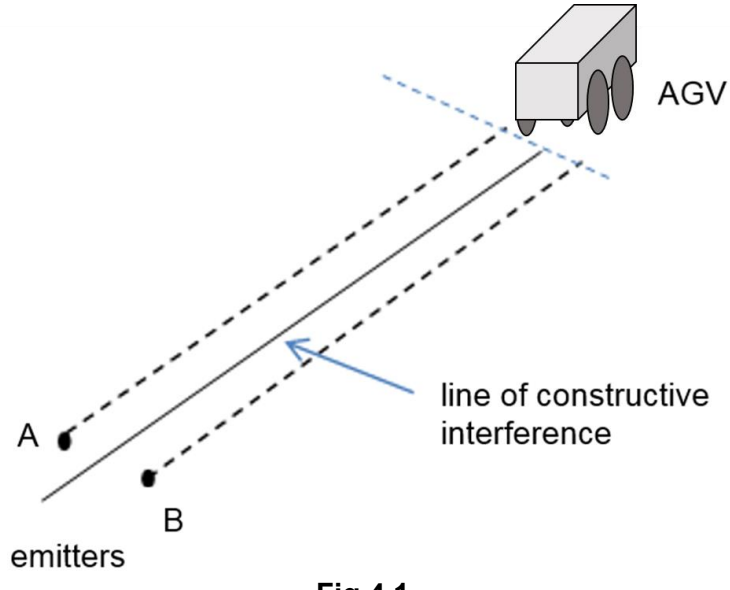

Fig 4.1

(i) State and explain one disadvantage in using this method for the AVG to align itself along the centre-line.

.....

.....

..... [2]

(ii) During one such operation, the AGV strays off the centre-line as shown in Fig. 4.2.

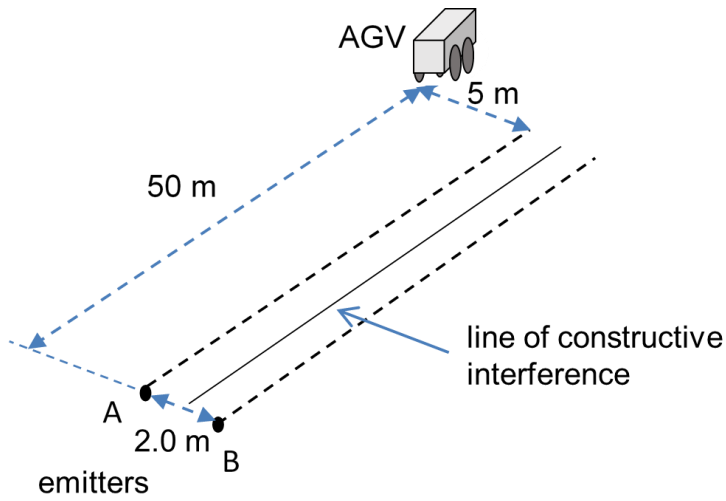

Fig 4.2

Fig. 4.3 shows the radio signals X and Y detected by the receiver on the AGV.

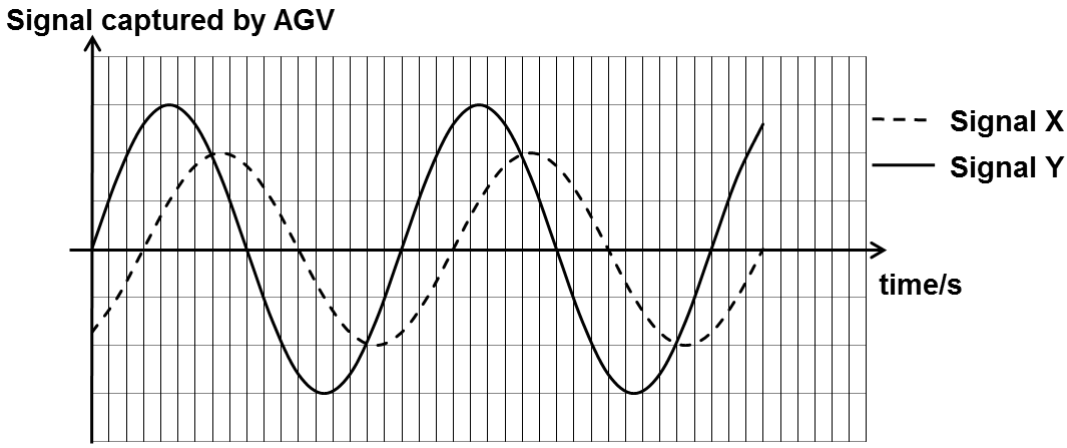

Fig 4.3

1. State and explain whether the source of signal X is from emitter A or B.

.....

.....

.....

..... [2]

2. State the phase difference between signals X and Y.

phase difference = ……. rad [1]

3. Assuming the path difference is less than one wavelength, determine frequency *f* of the radio wave used.

frequency *f* = ……. Hz [4]